

ELEC1100

Supplementary Notes for Project

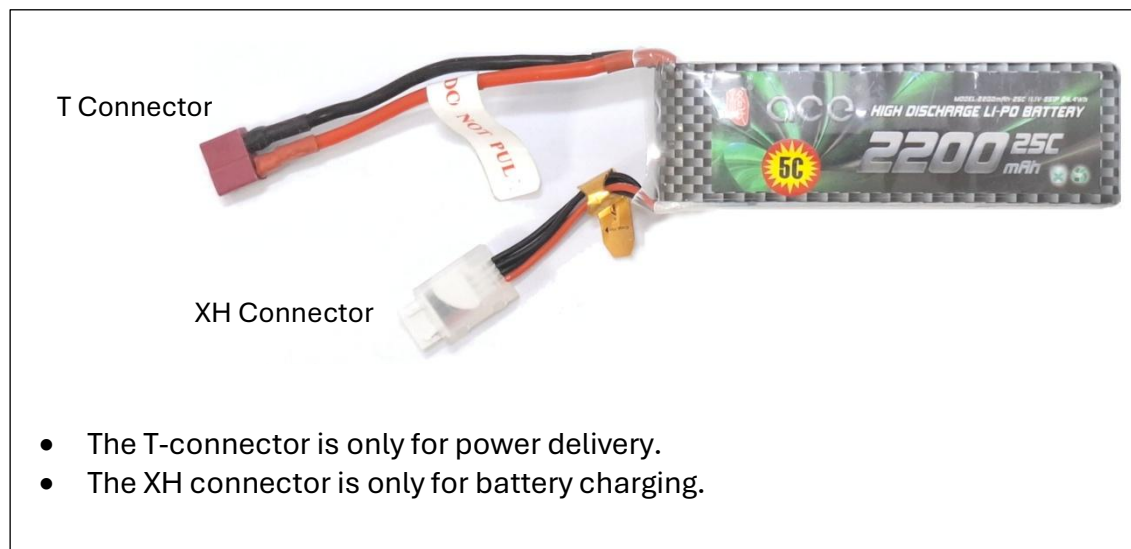
Contents

I.	Using battery and its accessories	2
A.	Battery.....	2
B.	Battery Charger.....	2
C.	Battery Monitor	5
D.	Using Battery with Breadboard	5
II.	Using LED with Line Sensor	6

I. Using battery and its accessories

A. Battery

Under normal conditions, the maximum output of the battery is ~12.6V after fully charged.



WARNING

DO NOT short any of the two terminals of connectors.
Otherwise, the battery would explode.

B. Battery Charger



Charging Steps

WARNING

DO NOT charge the battery without supervision.
Improper charging may result in damage to the charger or battery and could lead to hazardous conditions, including the risk of battery explosion.

- a) Connect the B5 charger to the AC adaptor.



- b) Connect the AC adaptor to the power socket.



- c) The LEDs for power and 1.2A charging current shall be lit. The battery type we used is LiPo.



- d) Connect the XH connector of the battery to the 3-cell charging port. Be careful with the XH connector orientation.



WARNING

Incorrect insertion will damage the XH connector and the balance port.
If the power LED does not lit, **DO NOT CONNECT** the battery to the charger.
Otherwise the charge would be damaged.

- e) When the battery is properly connected to the charger, the three green LEDs will flash. The battery is being charged.



- f) Once the battery is fully charged, all the green LEDs stop flash will remain steadily lit.

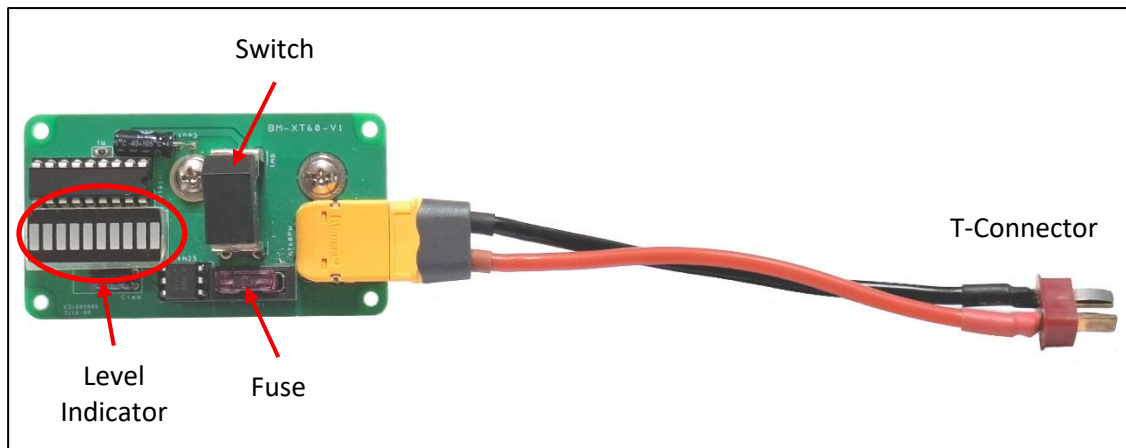


- g) Disconnect the battery from the charger.



C. Battery Monitor

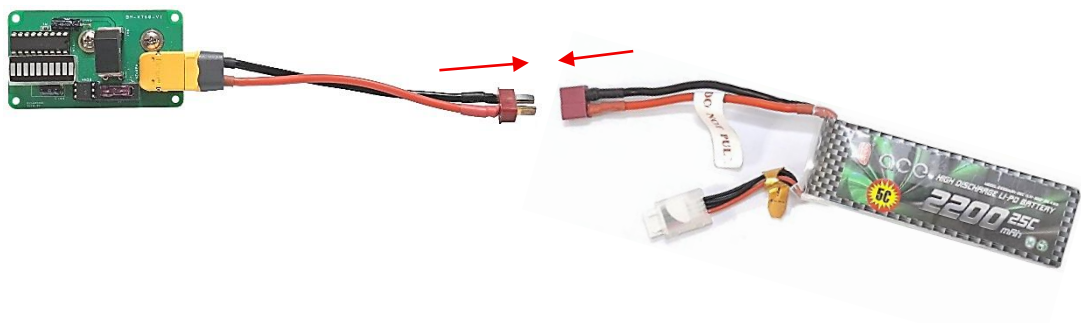
This is the battery monitor used in the project.



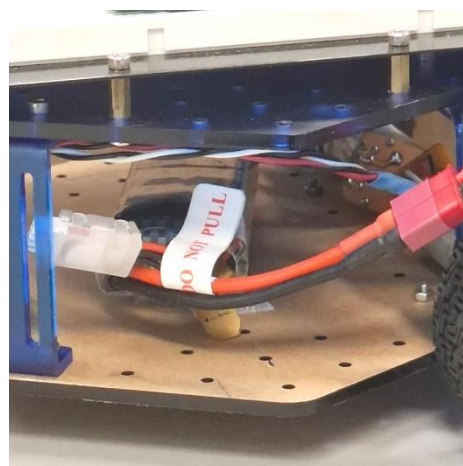
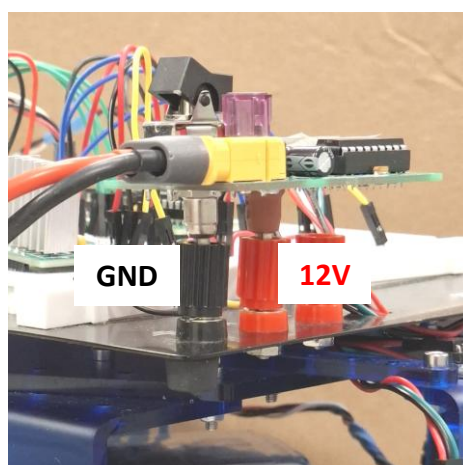
When the level indicator only shows red LED, it means your battery level is very low and you should charge the battery immediately.

D. Using Battery with Breadboard

1. Connect the battery to the battery monitor. Be sure to match the color of the cables on both sides.

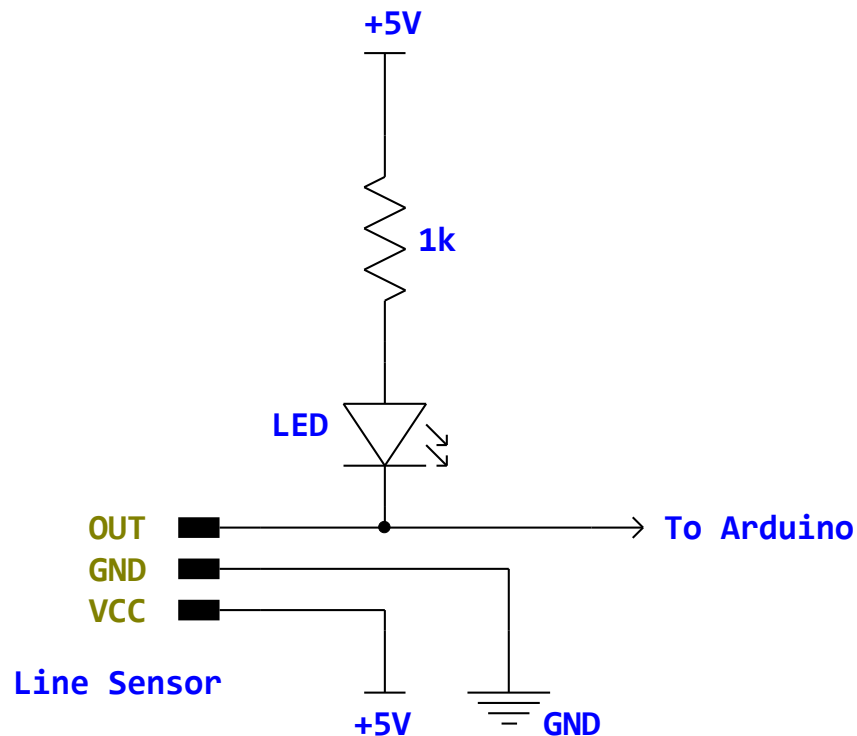


2. Plug the battery monitor into breadboard accordingly and place the battery into the car compartment.



II. Using LED with Line Sensor

You may want to use LED for easy observation of the status of the line sensor if it is on the white strip. The connection diagram below shows how to connect LED to the line sensor output on breadboard.



- 1) When the sensor is on top of the white strip, the sensor output level is low, the LED turns on.
- 2) When the sensor is on top of dark portion, the sensor output is high, the LED turns off.

A resistor MUST be used in series with LED in any cases.